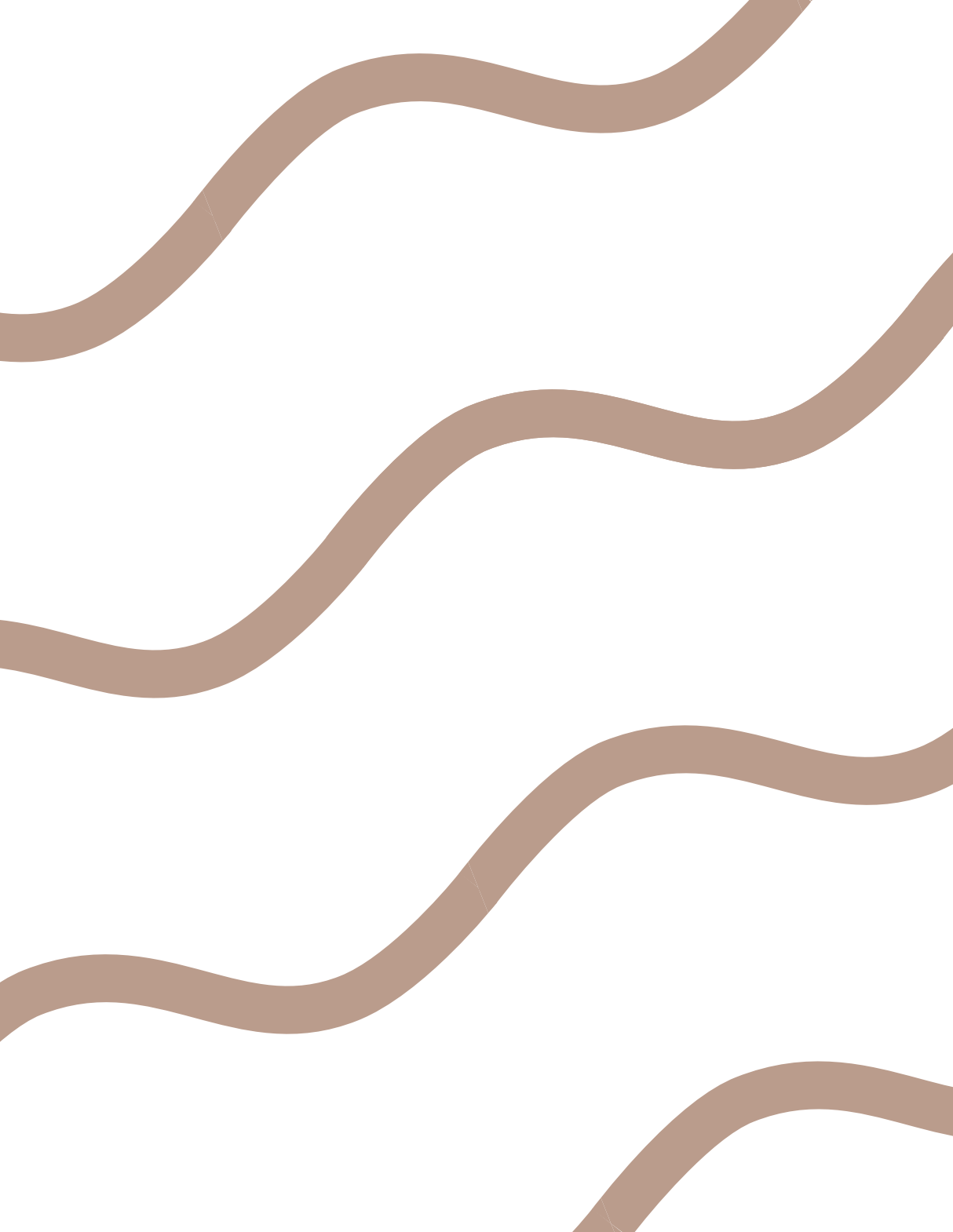




- data로부터 noise를 조금씩 더해가면서 data를 완전한 noise로 만드는 forward process (diffusion process)
- noise로부터 조금씩 복원해가면서 data를 만들어내는 reverse process

데이터에 noise가 더해진 상태의 latent variable

1. forward process
2. reverse process 학습

Reverse process : noise (X_T) $\rightarrow$ data (X_0)
 p_θ 를 활용해서 approximate

원본 데이터 $X_0 \sim q(X_0)$ (s) 순방향 X_0 에 T번의 step 노이즈 추가

$$q(X_t | X_{t-1}) = N(X_t; \sqrt{1-\beta_t} X_{t-1}, \beta_t I)$$

$\rightarrow X_t$ 는 X_{t-1} $\sqrt{1-\beta_t}$ 만큼 줄인 값 주변에, 분산이 β_t 인 가우시안 노이즈 추가

$$\alpha_t = 1 - \beta_t$$

$$q(X_t | X_{t-1}) = N(X_t; \sqrt{\alpha_t} X_{t-1}, (1-\alpha_t)I)$$

sampling

$$X_t = \underbrace{\sqrt{\alpha_t} X_{t-1}}_{\text{mean}} + \underbrace{\sqrt{1-\alpha_t}}_{\text{var}} \varepsilon_{t-1} \quad \text{where } \varepsilon_{t-1} \sim N(0, I) \quad \text{noise}$$

X_t (X_{t-2} 로 표현)

$$\begin{aligned} X_t &= \sqrt{\alpha_t} X_{t-1} + \sqrt{1-\alpha_t} \varepsilon_{t-1} \\ &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} X_{t-2} + \sqrt{1-\alpha_{t-1}} \varepsilon_{t-2}) + \sqrt{1-\alpha_t} \varepsilon_{t-1} \\ &= \sqrt{\alpha_t \alpha_{t-1}} X_{t-2} + \sqrt{\alpha_t (1-\alpha_{t-1})} \varepsilon_{t-2} + \sqrt{1-\alpha_t} \varepsilon_{t-1} \end{aligned}$$

$$\varepsilon \cdot \text{Var}(\sqrt{\alpha_t (1-\alpha_{t-1})} \varepsilon_{t-2}) = \alpha_t (1-\alpha_{t-1})$$

$$\cdot \text{Var}(\sqrt{1-\alpha_t} \varepsilon_{t-1}) = 1 - \alpha_t$$

$$\oplus 1 - \alpha_t \alpha_{t-1}$$

$$\sqrt{1-\alpha_t \alpha_{t-1}} \cdot \bar{\varepsilon} \sim N(0, I)$$

$$\therefore X_t = \sqrt{\alpha_t \alpha_{t-1}} X_{t-2} + \sqrt{1-\alpha_t \alpha_{t-1}} \bar{\varepsilon} \rightarrow \text{패턴 발견}$$

$$\bar{\alpha}_t = \alpha_1 * \alpha_2 * \dots * \alpha_t = \prod_{i=1}^t \alpha_i$$

$$X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1-\bar{\alpha}_t} \varepsilon \quad \varepsilon \sim N(0, I)$$

Title: _____

Name: _____

Date: _____

$$\int q(x_{1:T}|x_0) \log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} dx_{1:T}$$

$$KL(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t))$$

$$\text{mean} ; \frac{\sqrt{\alpha_{t-1}}}{1-\alpha_t} \beta_t \hat{x}_0 + \frac{\sqrt{\alpha_t(1-\alpha_{t-1})}}{1-\alpha_t} x_t + R_w$$

$$\log p_\theta(x_0)$$

$$\text{Var} ; \frac{1-\alpha_{t-1}}{1-\alpha_t} (1-\alpha_t)$$

P: reverse process

$$\text{mean} \quad \frac{\sqrt{\alpha_{t-1}}}{1-\alpha_t} \beta_t x_0 + \frac{\sqrt{\alpha_t(1-\alpha_{t-1})}}{1-\alpha_t} x_t$$

$$\left(\begin{aligned} \hat{x}_0 &= \frac{1}{\sqrt{\alpha_t}} (x_t - \sqrt{1-\alpha_t} \hat{\epsilon}_\theta(x_t, t)) \\ &\rightarrow \frac{\sqrt{\alpha_t}}{\sqrt{1-\alpha_t}} \beta + \frac{1}{\sqrt{\alpha_t}} \quad \text{approx.} \quad \|\epsilon - \hat{\epsilon}_0\| \end{aligned} \right)$$

$$\|\epsilon - \hat{\epsilon}_\theta(x_t, t)\|^2 \quad \epsilon \sim \mathcal{N}$$

$t = \dots$ $\rightarrow \sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t}$

$\hat{\epsilon}_t \quad t=1 \dots T$

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

$$= \dots \quad \frac{x_t - \sqrt{\alpha_t} x_0}{\sqrt{1-\alpha_t}} \quad \alpha_t^2 + (1-\alpha_{t-1} \alpha_t^2) (1-\alpha_t)$$

$$\begin{aligned} KL &= E_{q_0} [\log q_0 - \log p_\theta(x_{1:T})] \\ &= E_{q_0} \left[\sum_{t=1}^T KL(q(x_{t-1}|x_t, x_0) \| p_\theta(x_{t-1}|x_t)) \right. \\ &\quad \left. - \log p(x|x_0) \right] \end{aligned}$$

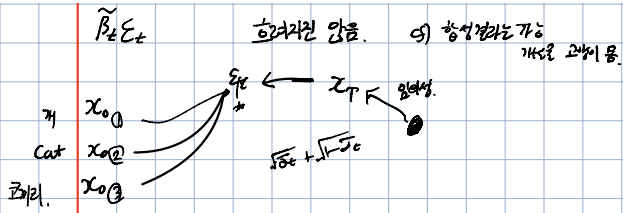
$$\frac{\|x_0 - \hat{x}_0\|^2}{\|\epsilon_t - \hat{\epsilon}_0(x_{t_0})\|^2}$$

$$\begin{aligned} x_{0:t} &= \sqrt{\alpha_{t-1}} \left(\frac{x_t - \sqrt{1-\alpha_t} \hat{\epsilon}_t}{\alpha_t} \right) \\ &\quad + \sqrt{1-\alpha_{t-1} \sigma_t^2} \hat{\epsilon}_t + \epsilon_t \end{aligned}$$

Title: _____

Name: _____

Date: _____



chapter 1. 특징 뽑을 가능하게

$P(y|x_t)$

생성할 때 $x_{t+1}|x_t$ 하듯이 conditional sampling?

$$p_\theta(x_t|x_{t+1}, y) \propto p_\theta(x_t, x_{t+1}, y)$$

$$= p_\theta(x_{t+1}) p_\theta(x_t|x_{t+1}) p(y|x_t)$$

$$\propto p_\theta(x_t|x_{t+1}) p(y|x_t)$$

타이머 연결

smoke를 사용하며 guide.

$$\pi_{x_t} \log p(y|x_t) \propto \hat{E}(x_t, y, t) - \hat{E}(x_t, \sigma_t)$$

$(1, 0, 0)$ $(0, 1, 0)$ $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

$\oplus$ Embedding.

$$dx_t = \mu(x_t, t) dt + \sigma(x_t, t) dz_t$$

normal 생성 가능

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Date : _____

Chapter 1 The Reinforcement Learning Problem

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